



RT-Pred: A web server for accurate, customized liquid chromatography retention time prediction of chemicals

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ABSTRACT

High-performance liquid chromatography (HPLC) together with mass spectrometry (MS) is routinely used to separate, identify and quantify chemicals. HPLC data also provides retention time (RT) which can be aligned with structural data. Recent developments in machine learning (ML) have improved our ability to predict RTs from known or postulated chemical structures, allowing RT data to be used more effectively in LC-MS-based compound identification. However, RT data is highly specific to each chromatographic method (CM) and hundreds of different CMs with interdependent parameters are used in separations. This has limited the application of ML-based RT predictions in compound identification. Here we introduce an easy-to-use RT prediction webserver (called RT-Pred) that predicts RTs for molecules across most chromatographic setups. RT-Pred not only supports its own in-house CM-specific RT predictors, it allows users to easily train a custom RT-Pred model using their own RT data on their own CM and to predict RTs with that custom model. RT-Pred also supports RT and compound searches against its own database of millions of predicted RTs spanning >40 different CMs. RT-Pred is also uniquely capable of accurately identifying compounds that will elute in the void volume or be retained on the column. Including this void/retained/eluted classifier significantly improves RT-Pred's performance. Tests indicate that RT-Pred had an average coefficient of determination (R^2) of 0.95 over 20 different CMs. Comparisons of RT-Pred against other RT predictors showed that RT-Pred achieved lower mean absolute errors and higher R^2 scores than any other published RT predictor. RT-Pred is freely available at <https://rtpred.ca>.

1. Introduction

Liquid chromatography-mass spectrometry (LC-MS) is a highly effective analytical technique that excels at separating complex mixtures into individual components based on their chemical properties and mass-to-charge ratios (m/z). Nearly all LC-MS methods use high-performance (HPLC) or ultra-high-performance (UPLC) systems in common modes such as reversed-phase liquid chromatography (RPLC) or hydrophilic interaction liquid chromatography (HILIC). These methods employ specially designed columns and stationary phases tailored to separate compounds via polarity-based interactions: RPLC separates analytes by hydrophobicity, while HILIC leverages hydrophilicity. Combined with high column pressures and optimized mobile phases, these systems achieve fast, clean, and reproducible separations. Because of the relatively short separation times (typically minutes to

tens of minutes), broad compatibility with diverse compound classes, and high separation power (e.g., resolution, peak capacity) of HPLC and UPLC methods, LC-MS is widely used in fields such as pharmacology, environmental analysis, natural product chemistry, lipidomics, exposomics and metabolomics to analyze complex chemical mixtures of unknown composition [1–3]. While the MS (or MS/MS) spectra measured by LC-MS methods are the primary data used to identify compounds, LC data can also provide chromatographic retention time (RT), which gives critical orthogonal information when aligned with structural data (through prediction models). In RPLC and HILIC, RT is predominantly governed by analyte polarity—hydrophobicity in RPLC and hydrophilicity in HILIC. However, molecular size, shape, and flexibility may also contribute secondary effects depending on the stationary phase and analyte class. As a result, combining RT with MS data allows RT information to be used as a “molecular filter”, efficiently

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narrowing down the pool of potential compound matches, a pool that might otherwise be much larger if MS data, alone, is used [4].

Unlike the situation with the Kovats retention index (RI), which is widely used in gas chromatography mass-spectrometry (GC-MS) [5], RT is not a universal measure that can be applied uniformly across all labs or across all LC platforms. This is because there are no known sets (yet) of internal standards, as there are with GC, that can be used to universally re-scale or re-calibrate a compound's RT measured via LC. Indeed, RTs are significantly influenced by the choice of the LC mobile and stationary phases, the mobile phase gradient, the flow rate, the LC column temperature, and the dimensions of the LC column, among other factors. The specific conditions or specific parameters for a given LC system are collectively called a chromatographic method (CM). Thus, a minor alteration in any CM has the potential to significantly change a compound's RT [4]. Therefore, to employ RT data in a compound identification workflow, many analytical laboratories maintain local libraries of RTs measured on their own collection of compounds using their own CMs [6,7]. These CMs are often quite unique, having been optimized by each lab to work with a given HPLC/UPLC instrument, a given column and a given set of analytes [8]. Not only is this RT-collection process time-consuming for individual labs, but it also leads to a considerable amount of inter-lab duplication. The result is a proverbial "Tower of Babel" consisting of thousands of different RTs collected for hundreds of identical or near-identical chemicals generated across multiple distinct CMs per lab. While RT data could, in theory, be shared across labs using the same CMs, small differences in setup—such as column batch variations, mobile phase additives, or gradient shifts—make direct comparisons difficult. As a result, most RT datasets are only useful within the lab that generated them. Hence, most measured RTs are useful for no more than a handful of labs and most experimentally measured RTs can only be used for identifying a few hundred well-known, easily purchased compounds [9]. This is because RTs for well-known commercial standard compounds are widely applicable as these compounds are routinely used as internal references across labs. Moreover, RTs for novel or less common analytes are rarely transferable due to methodological differences. Consequently, the use of RTs for identifying novel, rare or less accessible compounds or distinguishing between MS-ambiguous compounds is almost never done. In other words, the enormous potential of RTs for facilitating compound identification has not been fully realized [10].

Ideally, if RTs could be accurately predicted for a given CM and a given chemical structure (known or hypothesized), then the full potential of RTs in LC-MS analysis could be realized. Such a predictive system would allow millions of reference RTs, covering hundreds of thousands of compounds and thousands of CMs, to be computationally generated in a few CPU hours – instead of the decades of measurement that would likely be required if this were to be done experimentally [11]. The challenge, of course, is developing an accurate RT predictor. This has driven extensive research into computational approaches for RT prediction, evolving from early physicochemical models to modern deep learning techniques.

RT prediction has been explored since the 1980s [12], initially focusing on peptide RT prediction due to the emergence of LC-MS methods in proteomics and the increased use of HPLC for separating peptides prepared via solid-phase peptide synthesis. Thirty years later, in 2010, RT prediction began to be applied to metabolite and small molecule identification [4,13]. Initially these methods relied on linear regression approaches based on physicochemical variables (quantitative structure-retention relationships or QSRR) [4,14]. However, these methods were limited in their applicability to different CMs [15]. An important breakthrough occurred in 2015 with the development of PredRet [7] which used community-gathered RT data and a technique called "projection modeling" to pair CMs together with specific compounds (and their RTs). The method is remarkably accurate but is limited to predicting only a few hundred RTs for a given CM. Another important development in RT prediction came with the release of the 80,

038 molecule SMRT (small molecule retention time) dataset by the Siuzdak lab in 2019 [16]. The availability of such a large RT dataset, combined with other RT datasets collected via other CMs, has enabled the development of several machine learning (ML) methods for RT prediction [7,17]. Recent examples of some of the more accurate ML RT-prediction methods include Retip [18], RT-Transformer [19] and GNN-RT [20]. Others include MolCLR-Finetuned [21] and KAN-Basic [22] which are finetuned ML models that not only predict a wide range of molecular properties but can also be adapted for RT prediction. Most RT prediction methods employ deep learning methods such as transformers and graphical neural nets (GNNs) and many can achieve mean absolute errors (MAEs) between predicted and observed RTs of less than one minute under typical (15–20 min) chromatographic separations. Likewise, given a sufficiently large training set of compound structures and RTs (usually >200 pairs), these methods can predict RTs for almost any CM.

However, there are a number of shortcomings to many current RT predictors. First, they are not easily retrainable to adapt to a new CM. Indeed, most available RT predictors require a solid knowledge of programming and computer operating system management to install them and then successfully train and properly run an RT predictor. Second, most RT predictors are quite limited in the number of RTs that they can predict as users must format, assemble and even write specialized code to preprocess the input data needed by the program. Third, all RT predictors assume that every compound, no matter what CM is used, will elute. This does not accommodate the well-known fact that many compounds for any given CM will either be retained (a near infinite RT) or will come out in the void volume (an RT close to 0) [23]. Fourth, to the best of our knowledge, there are no existing RT prediction tools that can both search for compounds using predicted RT values and which can also find RT values for specific compounds through a compound search function. Finally, no RT predictor is available as a free, user-friendly web server.

To address these shortcomings, we have developed RT-Pred, an easy-to-use, freely available webserver for accurately predicting RTs. RT-Pred not only supports accurate RT prediction (for as few as one to as many as ~7000 molecules at a time), but it also allows users to easily retrain the predictor (via the web) to almost any CM using as few as 40 compound-RT pairs. These custom RT predictors can also be used for RT prediction (for 1 to 7000 molecules at a time). RT predictions for even more molecules can also be performed by simply submitting another job to the webserver. RT-Pred is uniquely capable of accurately predicting which compounds will elute, which will be retained, and which will appear in the void volume for any given CM. The RT-Pred website also contains a searchable collection of millions of predicted (pre-calculated) RTs generated for dozens of CMs corresponding to all compounds in the HMDB [24], FooDB [25] and the NP-MRD [26]. Detailed comparisons to other RT predictors indicate that RT-Pred is the most accurate and most capable RT predictor described to date. More details about RT-Pred, including how it was designed, tested and implemented, are given in the following pages.

2. Data and methods

2.1. Collecting RT data sets

To train, test and optimize RT-Pred, we assembled a number of experimentally measured RT datasets from diverse sources and a wide range of CMs. These include 20 RT datasets from the PredRet [7] collection of experimental RT data, including seven HILIC-based CMs (with 147 to 995 compounds), and 13 acidic C18-based reversed-phase (RP)-based CMs (with 70 to 1676 compounds). We also included 24 RT datasets collected on 467 plant compounds using 24 different RP-HPLC CMs from a comprehensive multi-laboratory study conducted by Low et al. [27]. In total, we compiled 44 RT datasets consisting of ~10,500 different compounds. A table showing the dataset distribution with CM

descriptions and references is available in the Supplementary Material (Table S1). To facilitate data storage and computational analysis, all chemical structures in the RT datasets were converted to a simplified molecular-input line-entry system (SMILES) format [28]. For those compounds not already in a SMILES format but available in an InChI key or an InChI string, we developed our own InChI-to-isomeric-SMILES converter using RDKit [29] and ChemSpider [30]. A number of structure and InChI errors were identified during the format conversion process. For compounds with incorrect InChIs, we searched by compound names and cross-checked with similar compounds to obtain the correct structure representation. After generating the SMILES strings, we manually verified each SMILES string by visualizing the structures using chemical drawing software such as ChemAxon's Marvin tools [31] and comparing them with known molecular structures in established databases like PubChem [32], and checking for consistency, particularly with respect to stereochemistry and isomer designation. This manual

verification process ensured the overall accuracy and reliability of all chemical structures used in the RT datasets.

2.2. Identifying and clustering RT groups

As noted earlier, regardless of the CM chosen, any complex chemical mixture will likely have some compounds that elute in the void volume (RT near 0), other compounds that elute throughout the chromatographic run time (well-behaved RT) and some compounds that are essentially retained in the column or elute long after the bulk of the compounds have already been eluted (retained or RT near infinity). The quality of an RT predictor can be greatly improved if it is trained on compounds that are well-behaved and if the compounds eluting in the void volume (known as void compounds hereafter) and retained compounds are left out of the regression model. Likewise, developing a separate predictor that classifies molecules as likely to be void, eluted or

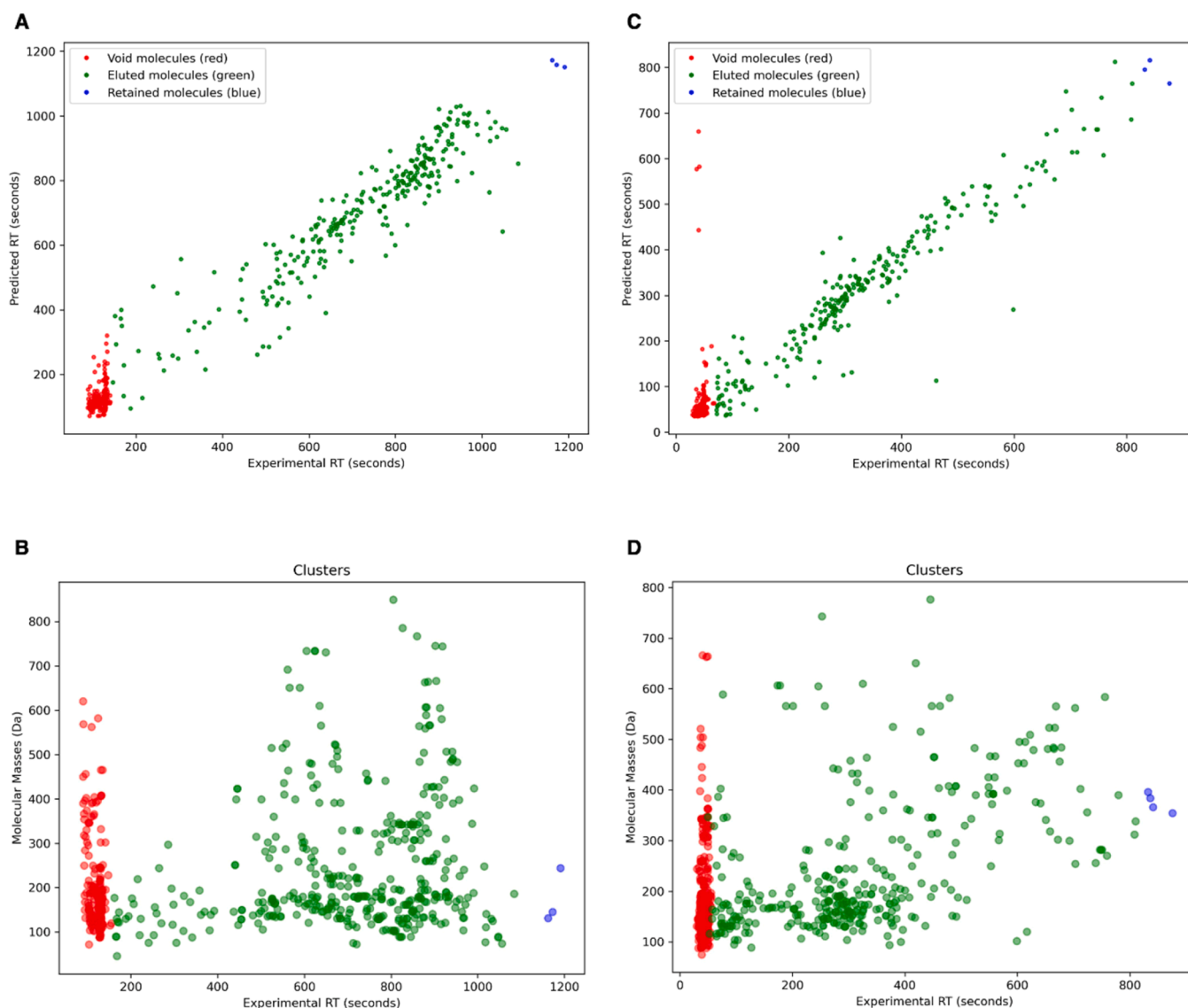


Fig. 1. An illustration of the relationship between predicted and experimental retention times (RT) and molecular masses for HILIC_BDD 2 chromatographic method (Merck SeQuant ZIC-HILIC using ACN 0.1% formic acid and water with 16 mM ammonium formate) given in (A and B) and for BDD_C18 (Reverse Phase in Hypersil Gold 1.9 μ m C18 with Water:ACN 0.1% and formic acid eluent) given in (C and D). Void compounds (red), which elute in the first 5–10% of an LC run, form a vertical cluster near the origin with low RTs (retention times) and molecular mass (in Daltons or Da). Retained compounds (blue), appearing after a long pause in the later stages of the LC run, cluster horizontally in the top-right corner with high RT and mass values. Eluted compounds (green) span a broad range of RT and masses between the void and retained categories. (A) and (C) highlight the correlation between predicted and experimental RT, while (B) and (D) show corresponding molecular mass clustering, validating the GMM-based categorization developed in custom models. Abbreviations: ACN - acetonitrile; HILIC - hydrophilic interaction liquid chromatography.

retained, prior to RT prediction, would be expected to improve the quality of an RT predictor. Therefore, we grouped all compounds in our RT datasets as void (class = 0), eluted (class = 1) and retained (class = 2). As a general rule, void compounds appear in the first 0–5% of an LC run while retained compounds appear after a long pause in the last eluted compounds of an LC run. When comparing predicted RT values (on the Y-axis) with observed RT values (on the X-axis), the void compounds show as a vertical cluster near the origin while the retained compounds display has a horizontal cluster on the upper corner, distinct from the cluster corresponding to eluted compounds (see Fig. 1).

To identify void and retained compounds, we ran a modified version of GNN-RT [20] against all 44 RT datasets and manually identified clusters of metabolites or compounds that exhibited the features seen in Fig. 1 where Panel A and B were prepared using the HILIC BDD_2 CM dataset and Panel C and D were prepared using the RP BDD_C18 CM dataset. Panel A and B had a labeled training and testing dataset that consisted of an average of 73 void (class = 0) compounds, 4 retained (class = 2) compounds and 428 eluted compounds (class = 1) for an average-sized dataset of 505 compounds for a HILIC based CM. Whereas, Panel C and D had a labeled training and testing dataset that consisted of an average of 345 void (class = 0) compounds, 3 retained (class = 2) compounds and 320 eluted compounds (class = 1) for a RP based CM with 671 compounds. This was the base method used to cluster all the existing CMs in our server (which took advantage of the domain knowledge of the general rule stated above). We also plotted the molecular mass of the compounds clustering within the void, eluted and retained categories (Fig. 1, Panel C and D) against their experimental RT. Compounds that varied in weight from 50–1200 Da were generally found within the void and eluted clusters while only three to four compounds were seen in the retained group with masses less than 400 Da. This demonstrates that other molecular descriptors besides MW, such as polarity, hydrophobicity, charge and other structural features, contribute to retention behavior in HILIC and RPLC.

2.3. Predicting RT groups from molecular structure

After the RT datasets had been clustered into void, eluted, and retained categories, we developed a separate ML tool to predict which RT group a given molecule would be classified based on its chemical structure and a given CM. The ML method we chose was XGBoost or extreme gradient boosting [33]. XGBoost is a decision tree-based learning method known for its efficient classifications and its ability to achieve accurate predictions requiring only a few seconds even when handling large datasets [34]. XGBoost has been shown to outperform support vector machines (SVM) and ensemble random forests (RF) when applied to chemical property prediction tasks using molecular fingerprints [35] and other molecular descriptors derived from SMILES strings [36]. XGBoost works by combining the predictions of multiple decision trees, where each new tree is built to correct the errors of the preceding trees [37]. This method focuses on minimizing the loss function using Newton-Raphson optimization. To train our XGBoost RT-group classifier (RTGC), we used RD-kit to extract structural features from the SMILES strings, including extended connectivity fingerprints [38] and molecular size descriptors. To improve model efficiency and to reduce dimensionality (and reduce overfitting), we also employed principal component analysis (PCA) [39] using the scikit-learn library [40]. PCA can be an effective tool for feature selection in machine learning by reducing dimensionality, limiting redundancy and retaining only the most significant information from a dataset. Our dataset contained 2049 features per compound extracted from fingerprints and size descriptors and we applied PCA to retain 95% of the data variance. We began by fitting the PCA results to our training sets features, sampling 30% of our data without specifying the number of components. By computing the cumulative variance explained by each component, we determined the smallest number of components needed to reach 95% variance, which ranged from 19 to 324 features per compound depending on the dataset.

We then applied PCA using that optimal number of components to the training and testing data to ensure consistency. This dimensional reduction step helped to select the most informative structural features from the extracted data before being fed into the XGBoost model for group classification. Finally, we used the XGBClassifier model (available in the XGBoost library), which is specifically designed for classification tasks, to perform the RT group classification. The RTGC is a trained version of this model and is unique for every CM. It was trained on 90% of the dataset (randomly selected with about 505 compounds per CM) and tested on a holdout set of 10% compounds.

2.4. Design, training and testing of the RT-Pred algorithm

We specifically designed the RT-Pred algorithm to predict RT data for class-1 (eluted) compounds. We chose the GNN-RT model [20] as our starting model, based on its excellent RT prediction performance, its source-code stability and the fact that it had been previously trained and refined on the METLIN SMRT dataset – a dataset which contains RT information for 80,038 small molecules [16]. We also chose GNN-RT as our base model because it was built using a graphical neural network design. Graphical neural networks (GNNs) [41] are a type of deep learning model designed to operate on graph-structured data, where entities (nodes) and their relationships (edges) are represented as graphs. In the context of molecular property prediction, GNNs excel because they can naturally represent molecules as graphs, with atoms as nodes and chemical bonds as edges. This structure allows GNNs to effectively capture both local (atomic) and global (molecular) interactions, making them highly effective at learning complex patterns and relationships within molecular structures, leading to accurate predictions of properties, such as RT [42].

While RT-Pred borrows code from the open-access GNN-RT [20], we significantly modified GNN-RT to increase its accuracy and make it far more robust in terms of RT predictions. Specifically, we limited the training, testing and validation of the model to eluted (class 1) compounds only to avoid model deterioration. We also incorporated stratified sampling which divides the dataset into “strata” based on chemical properties. By sampling each stratum proportionally, this technique ensures diverse and representative samples [43] and reduces bias, as the full range of RTs becomes evenly distributed across the training and test sets. For feature generation, we used RD-Kit to extract molecular fingerprints via the Weisfeiler-Lehman algorithm [44], along with other molecular features such as molecular size, chirality, etc. To identify the most relevant descriptors, we employed LASSO regression for feature selection. LASSO regression applies regularization to rank and selects the top features by shrinking less informative feature coefficients to zero, resulting in a more precise and reliable model. As with all GNNs, the GNN component of RT-Pred captures bond information between atoms and generates node IDs for each atom based on its neighboring nodes and edges. Additionally, we created an adjacency matrix to represent the connectivity graph of each input molecule. Molecular embedding vectors were generated using multiple layers of the GNN, which receive the aforementioned features extracted from each SMILES input. These vectors were then processed to create meaningful molecule representations which were fed through fully connected layers to predict RTs.

Using the eluted molecule set (identified from RTGC) from the SMRT database [16], which consisted of 77,864 eluted molecules out of 80,038, along with the algorithmic modification described earlier, we retrained RT-Pred to create a new baseline model that was both different and more accurate than GNN-RT. To allow the RT-Pred model to handle the 44 training/testing CMs with the 44 sets of RT data, we incorporated similar transfer learning (TL) or fine-tuning techniques as found with GNN-RT [20]. With TL, the baseline RT-Pred model can be trained on a smaller RT dataset. Certain layers in the GNN are frozen in the TL process—typically the initial layers that capture fundamental graph features—by preventing their weights from being updated. This means

only the final layers, which adapt the model to the specific task, are trained further. Freezing stabilizes the learned general features and reduces overfitting, especially when the training dataset is small. After TL, the GNN retains useful pre-trained knowledge while being tailored to the new task. The hyperparameters selected by us to train RT-Pred for both the baseline and the TL based model are listed in the Supplementary Material (Table S2). In RT-Pred's training phase, we implemented batch learning and 10-fold cross-validation to ensure robust and efficient model performance. We used a cross-validation approach by dividing the training data (excluding the testing set) into 10 equal folds. For each fold, one subset was used as the validation set, while the remaining data was used for training. This ensured that each data point served as a validation sample exactly once throughout the entire process. During training within each fold, the model also processed data in mini batches of 32 samples, calculating the loss by comparing predictions to actual values while applying regularization to prevent overfitting. The validation set allowed us to evaluate model performance on unseen data in each iteration, guiding hyperparameter tuning and model improvements. Mean absolute error (MAE) was used as the evaluation parameter to update the model's parameters and minimize the loss to improve prediction performance. Moreover, we identified the optimal number of training epochs for each model through trial runs and early stopping, which halts training when validation performance plateaus, thus preventing overfitting. The models typically reached optimal performance between 250 and 350 epochs, although the exact number varied slightly based on dataset size and complexity. After completing all training folds, we averaged the training and validation metrics to obtain a comprehensive performance measure before training a single, final model on the entire training dataset. The final model was then evaluated on the held-out test set comprising 10% of the dataset after stratified sampling.

While the training process described above works well for an average sized RT dataset of ~500 compounds, we often encountered much smaller RT datasets that were tailored to specific compounds or research questions with a limited number of measured analytes. A small RT dataset generally contains fewer than 100–200 unique analytes or compounds. Training a ML model on such a small dataset can be challenging, as it lacks sufficient diversity to generalize well, resulting in models that are prone to overfitting – capturing noise rather than true patterns [45]. Examples of these small CM/RT datasets can be found in the 24 gradient-based CMs published by Low et al. [27], which consists of experimental RT values for plant-derived compounds. Building RT models for these CMs was difficult as each individual dataset was relatively small, ranging from 29 to 109 compounds. To overcome the limitations of the smaller datasets, we first tried incorporating Bagging (Bootstrap Aggregating) [46] to enhance the performance and stability of generated RT predictive models. However, since Bagging involves creating multiple subsets of the original dataset, the small size of the datasets provided very few elements upon which we could train the model. Hence, we took a different approach, implementing leave-2-out (L2O) [47] testing for these models while training on the remaining $n-2$ data points. L2O testing is particularly useful when the dataset is small, and where every data point contributes significantly to the model's understanding. It provides a more robust evaluation of the model's generalizability by considering various combinations of training and testing instances. The process is computationally expensive as it involves training and testing the model multiple times, but it can offer a more reliable estimate of the model's performance, especially when dealing with limited data. Consequently, we started iteratively training models on all data points except two, repeated recursively for a total of 10 models per CM. This process led to a cumulative testing set comprising 20 data points, and statistical calculations based on these results were then presented. The final models used in the server were trained on the complete datasets.

Despite the baseline model for RT-Pred being made from the RP-centric SMRT dataset [16], applying the same TL technique to the

HILIC datasets also yielded excellent results. For retained and void molecules identified by the RTGC algorithm, RT-Pred calculates the average RT of all the molecules in each retained and void cluster and then assigns the average RT value as the predicted RT for each of the retained or void molecules identified in those groups.

2.5. Modifying RT-Pred to accept new CMs

A key functionality offered by the RT-Pred server is the ability to generate RT predictors for custom CMs. Manually labelling new datasets, to void, eluted and retained clusters, in real time is not feasible. As a result, we generalized our RT classification process, without using a hard RT cutoff, for all CMs to classify compounds into class 0, 1 or 2. We used Gaussian mixture modeling (GMM) [48] to perform this clustering and verified its performance by testing on the manually labelled 44 datasets. This testing confirmed that our GMM can perform very accurate clustering to produce a labelled dataset. GMM, a soft clustering technique often used in unsupervised learning, estimates the probability that a given data point belongs to a cluster. GMM assumes all data points are generated from a mixture of a finite number of Gaussian distributions with unknown characteristics, which is particularly useful in clustering complex datasets in chemistry [49,50]. Clusters are usually initiated via a random starting point and GMMs then adjust the average values and standard deviations within these clusters (known as radial and angular clusters), using an expectation-maximization algorithm, to optimize clusters [51,52]. In our case, the radial clusters represent groups of molecules with similar RT properties, while angular clusters represent groups of molecules sharing similar structural features, such as the molecular fingerprint extracted from the SMILES string.

To create our feature set, we extracted structural features consisting of extended connectivity fingerprints, molecular size (calculated by converting a SMILES string to a molecule, adding hydrogen atoms, and counting the atoms), atomic adjacency, and bond-edge information from each SMILES string in a RT dataset. The structural features were then combined with the corresponding RTs to create the final feature set to feed into the GMM, developed using the gaussian mixture class from the sklearn.mixture module in the Scikit-Learn library [40]. The GMM dynamically adapts to CM variability (e.g., gradient delays, column chemistry) by learning probabilistic boundaries from labeled data in the SMRT dataset and then feature transferring to the given CM dataset. We tracked the silhouette scores (a measure of how compact and well separated the clusters are, with a score nearer to one being better) to optimize our GMM [53]. It is important to remember that GMMs can use both data relationships [54] and labeled data to improve grouping. Indeed, GMMs partially learn on the feature representation from the labeled data and partially learn on the cluster structure using both labeled and unlabeled data. When we tested the GMM's performance on our RT-classified datasets, we found that it accurately identified compounds within their RT clusters, achieving an average accuracy of 92%. Hence, our final GMM was able to autonomously cluster any RT dataset into void, eluted, and retained categories without needing labeled data.

Once the user-provided dataset is labeled by the GMM, it is then used by our XGBoost-based RTGC model to predict RT classes or groups. This enables RT-Pred to correctly label new compounds as void, eluted, or retained for the custom CM without requiring their RT. Moreover, the labeled "eluted" compounds from the dataset can then be used to train the GNN component of RT-Pred, which then predicts RTs for new compounds in the custom CM, following the procedure described earlier. Fig. 2 shows the workflow that leads to the final RT prediction in both existing and new CMs.

2.6. Development of the RT-Pred webserver

To make the RT-Pred program and other RT classification tasks more accessible and easier to use, we developed the RT-Pred webserver. This webserver supports five functions: 1) predicting RTs for one or more

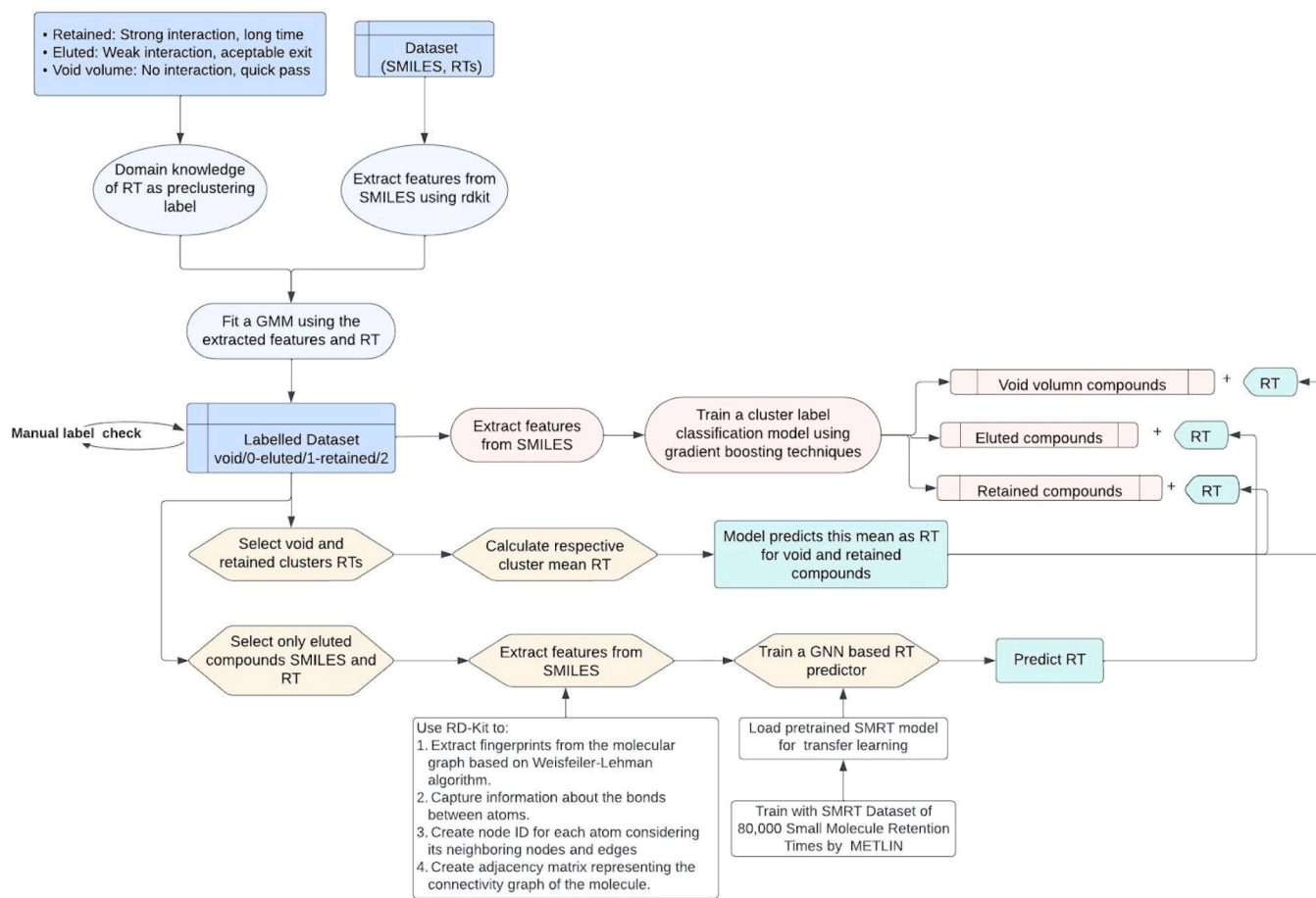


Fig. 2. The workflow for predicting retention times (RTs) and building custom models using ML for different chromatographic methods (CMs). Specified features are extracted in each step from SMILES strings (sent as inputs) with RDKit. Domain knowledge is used for manual clustering of molecules, into void, eluted and retained, for each of the given 44 CMs, while gaussian mixture modelling (GMM) is used for clustering molecules for new custom models. An XGBoost classifier is trained to label void, eluted and retained compounds and a graph neural network (GNN) handles RT prediction for eluted compounds using transfer learning with the SMRT dataset trained model.

input molecules using one of RT-Pred's pre-trained CMs; 2) creating custom RT predictors through user-supplied CM data; 3) predicting RTs for one or more input molecules using a custom CM and custom RT predictor; 4) searching molecules (via SMILES strings) and obtaining predicted RTs for multiple pretrained CMs; 5) searching an RT range for a given CM and obtaining lists of matching molecules. The latter two functions exploit RT-Pred's large database of pre-calculated RTs for multiple CMs of well-known metabolites and natural products.

The RT-Pred webserver was built using standardized web frameworks and caching systems previously developed in our lab [26]. More specifically, the frontend was built in Python with the Django framework. This was done using Django's built-in Object-Relational Mapping (ORM) for efficient development, along with HTML, JavaScript, and CSS. The Django Model-View-Template (MVT) architecture was used to enhance code organization making it easier to manage, test, and scale the web application to deliver a seamless user experience. The framework also has built-in security measures to protect against common web vulnerabilities. We repurposed some of the CSS and JavaScript design files from our Natural Product Magnetic Resonance Database (NP-MRD) [26] webserver to create a consistent user interface "look" for the RT-Pred server.

The RT-Pred webserver follows an open-access design, meaning that users do not need to login, even when creating a custom RT prediction model. Data are stored in a MySQL database and each user-assigned CM is saved in a dedicated repository under the assigned name for one-time session access. Users can make multiple RT predictions with their

custom RT model during each session. Through RT-Pred's interface, users may also request to have their custom RT prediction models made publicly available through a newly added CM category. To avoid misuse or abuse, these deposited RT prediction models will be tested by the RT-Pred team for accuracy, relevance and completeness before final public release.

3. Results and discussion

3.1. Training data anomaly detection and correction

While developing and testing different RT-Pred prediction models, we identified a few anomalies in both the molecular structures and RTs within selected datasets. For example, phloridzin, in the AjsUoB dataset from PredRet [7], had an RT of 75 ss (1.25 mins), although RT values for the same molecule in MassBank data are much higher, between 4.7 and 5.2 mins [55]. Consequently, we excluded phloridzin from the training set (interestingly, phloridzin's RT was later predicted by RT-Pred as 286 ss or 4.76 mins, which aligns with the MassBank result). Altogether we identified 43 molecule/RT pairs overall that had to be excluded, or which were manually corrected. To identify other discrepancies, we calculated pairwise Tanimoto scores of every compound in our dataset. Reported RTs of closely matched compounds (with high Tanimoto scores, ≥ 0.9) were compared and analyzed, with the assumption that compounds with very similar molecular structure would be expected to have similar RTs for the same CM [56,57]. Molecules that exhibited

significant RT discrepancies were manually checked and verified against other established RT databases. This process identified a small number of erroneous RTs that were subsequently corrected.

Additionally, we encountered errors related to molecules with neutral tetra-coordinated nitrogen atoms (neutral nitrogen should only form three bonds) when converting structures from InChI to SMILES using RDKit. This led to errors in representing nitrogen's valency. The number of compounds having this issue varied and was dependent of the chemical composition of the dataset. To resolve this issue, we reviewed and manually corrected those molecules containing this erroneous nitrogen valency in each dataset, ensuring proper adherence to valency rules and accurate chemical structure representation from RDKit.

3.2. Evaluating RT prediction models for 44 different CMs

As noted earlier, void and retained compounds can substantially diminish the performance of an RT predictor as nearly all RT predictors assume compounds behave linearly, as opposed to categorically. Identifying the void and retained compounds and removing them from our regression models would be expected to improve the quality of any RT predictor. Our XGBoost RT-group classifier (RTGC) accurately identified void and retained molecules in the CM datasets. Table 1 lists the number of void volume and retained molecules detected in each CM. On average, 74 (range 2–345) void volume molecules and 5 (range 1–17) retained molecules were detected per CM. Of all the datasets, the AjsUoB, the RIKEN, the Hilic_BDD_2 and the BDD_C18 datasets had the highest percentage of void volume + retained molecules (22.3%, 33%, 31% and 52.3%, respectively). Interestingly, the number of identified void + retained compounds was independent of the dataset size (large or small) or column type (HILIC or RP). Fig. 3, which shows the correlation between predicted and experimental RTs, with and without RTGC filtering, illustrates an example of the impact of applying RTGC filtering on the BDD_C18 dataset. Using RTGC significantly improved model performance as the coefficient of determination (R^2) increased from 0.92 to 0.97. Likewise, the MAE was reduced from 0.4 min to 0.24 min and the RMSE (Root Mean Squared Error) was reduced from 0.92 min to 0.59 min. These findings highlight the effectiveness of RTGC in improving RT

predictions. The performance of the RTGC algorithm is detailed in Table 1, specifically in the last column under RTGC Classification Accuracy. This column reports accuracy based on a randomly selected 10% of testing compounds from each dataset. Notably, the RTGC algorithm successfully predicts all three compound states when trained on a specific CM, regardless of whether the CM employs RP or HILIC columns. Note that for the plant bioactive compounds in the 24 RP chromatographic datasets (Table 2), we did not classify void, eluted and retained molecules because the datasets were too small. Additionally, the dataset names used in the tables (e.g., AjsUoB, Fem_Lipids, CS1, CS2 or chromatographic systems) are abbreviations for different CMs and are the exact names given to them from the sources they are collected. More details about these CMs and what the abbreviations correspond to are included in Table S1 of the Supplementary Materials, where full descriptions, including dataset origins, are available for cross-referencing.

We trained all CM datasets with RT-Pred and used 10% of the large datasets as holdout sets from PredRet (Table 1) and the L2O ensemble test set as a holdout set for smaller datasets (Table 2) to evaluate RT-Pred's performance. For the RP and HILIC datasets (20 CMs in Table 1), an average R^2 of 0.95 (range 0.85 - 0.99) for the training datasets and 0.91 (range 0.8 - 0.98) for the testing datasets was seen, demonstrating RT-Pred's excellent predictive capability. The training data had an average MAE of 0.46 mins, a MedAE of 0.22 mins, and an RMSE of 0.78 mins. However, the test set had little higher values with an average MAE of 0.73 mins, a MedAE of 0.36 mins, and an RMSE of 1.19 mins, indicating some loss of accuracy on unseen data. Top CM performers, like AjsUoB and BDD_C18, had very low error metrics (953 compounds, with $R^2 = 0.98$, MAE=0.16 min for the training set and $R^2 = 0.97$, MAE=0.17 min for the testing set). Other datasets such as Eawag_XBridgeC18 (363 compounds, training MAE=0.51 min, testing MAE =0.9 min) and UFZ_Phenomenex (394 compounds, training MAE=0.78 min, testing MAE=1.5 min), showed larger performance drops from training to testing. This may be due to the greater chemical diversity of these CM datasets. Despite challenges posed by certain models, the RT-Pred program shows good generalizability. However, the degraded predictive performance seen in smaller datasets or datasets with more chemically diverse structures simply affirms the general ML rule that "more data

Table 1

Performance evaluation and error calculations (in minutes) of RT-Pred on chromatographic methods (CMs) from PredRet.

CM Name	Size of Dataset	Training				Testing*				Number of compounds			RTGC Classification Accuracy
		R^2	MAE	MedAE	RMSE	R^2	MAE	MedAE	RMSE	Void Volume	Retained	% Void volume + retained	
AjsUoB	953	0.98	0.16	0.00	0.33	0.97	0.17	0.00	0.41	208	5	22.35	0.97
Fem_Long	418	0.99	1.31	0.68	2.16	0.98	1.66	0.51	2.69	84	6	21.53	0.96
Fem_Lipids	70	0.98	0.61	0.10	0.98	0.95	1.09	1.15	1.5	9	4	18.57	0.97
Life_Old	193	0.95	0.15	0.10	0.23	0.92	0.18	0.06	0.31	32	1	17.10	0.89
Life_New	184	0.87	0.32	0.09	0.54	0.88	0.34	0.18	0.50	31	1	17.39	0.95
RIKEN	470	0.85	0.50	0.09	1.15	0.91	0.40	0.00	0.98	154	1	32.98	0.87
Eawag_XBridgeC18	363	0.96	0.51	0.30	0.79	0.90	0.90	0.76	1.26	17	2	5.23	0.96
BfG_Nts_RP1	912	0.99	0.28	0.19	0.41	0.93	0.78	0.44	1.11	7	3	1.10	1.00
Hilic_BDD_2	592	0.97	0.51	0.14	1.03	0.92	0.67	0.00	1.56	180	3	30.91	0.91
UniToyama_Atlantis	143	0.97	0.68	0.52	0.92	0.86	1.52	0.54	2.38	2	2	2.80	0.94
BDD_C18	671	0.98	0.19	0.00	0.48	0.97	0.20	0.00	0.48	345	6	52.31	0.93
RPMMPDA	1676	0.98	0.18	0.12	0.27	0.93	0.37	0.24	0.58	75	7	4.89	0.98
SNU_RIKEN_POS	672	0.96	0.29	0.21	0.43	0.94	0.35	0.26	0.52	36	10	6.85	0.89
UFZ_Phenomenex	394	0.97	0.78	0.37	1.27	0.91	1.50	0.49	2.32	12	6	4.57	0.97
MTBLS87	147	0.90	0.85	0.49	1.37	0.80	1.11	0.78	1.78	6	2	5.44	1.00
KI_GIAR_zic_Hilic_pH2.7	538	0.97	0.32	0.15	0.52	0.90	0.61	0.20	1.06	48	5	9.85	0.93
Meister_zic-pHilic	295	0.89	0.56	0.32	0.89	0.87	0.75	0.54	1.02	32	5	12.54	0.90
AjsTestF	995	0.96	0.29	0.10	0.56	0.91	0.43	0.06	0.86	183	17	20.10	0.85
LJM_TEST	349	0.97	0.45	0.16	0.80	0.89	1.11	0.69	1.74	7	3	2.87	1.00
IPB_Halle	82	0.96	0.34	0.16	0.55	0.93	0.48	0.35	0.71	6	1	8.54	1.00
Average ± Standard Deviation	505 ± 616	0.95 ± 0.01	0.46 ± 0.13	0.22 ± 0.11	0.78 ± 0.16	0.91 ± 0.03	0.73 ± 0.22	0.36 ± 0.25	1.19 ± 0.21	73.7 ± 90.3	4.5 ± 3.71	14.90 ± 12.86	0.94 ± 0.04

Abbreviations: MedAE - Median Absolute Error, MAE - Mean Absolute Error, and RMSE - Root Mean Squared Error

* 10% of data from the datasets collected from PredRet were held out for testing.

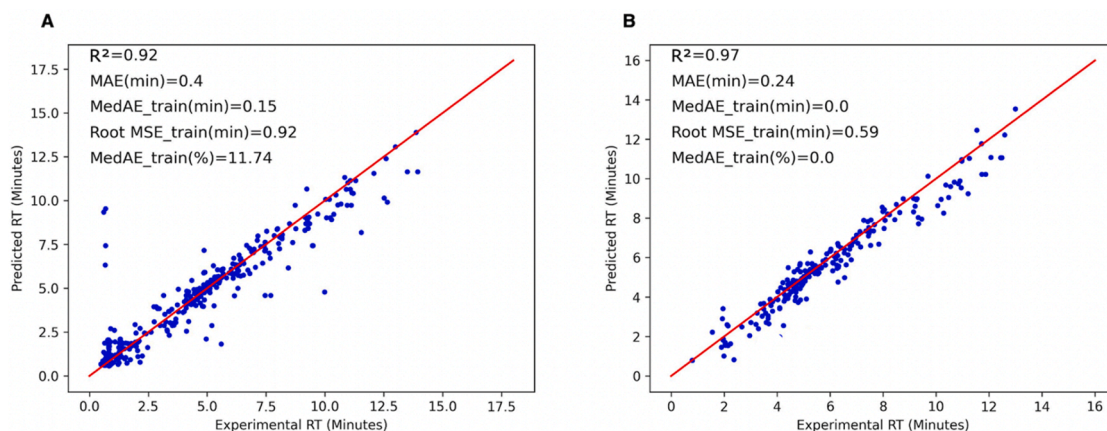


Fig. 3. Correlation plots showing the predicted versus experimental retention times (RTs, in minutes) predicted by RT-Pred using the training dataset BDD_C18 (Reverse Phase in Hypersil Gold 1.9 μ m C18 with Water:ACN 0.1% and formic acid eluent). Panel A shows the results without the 3-state classifier and panel B shows the results after incorporating the classifier for RT prediction of void, eluted and retained molecules. This approach significantly improved model performance as R^2 increased from 0.92 to 0.97, and MedAE (Median Absolute Error), and RMSE (Root Mean Squared Error) were reduced. Abbreviation: ACN - acetonitrile.

Table 2

Performance evaluation and error calculations (in minutes) of RT-Pred on the smaller (plant bioactive compound) datasets in the 24 RP chromatographic methods (CM) tested and trained using Leave-2-Out method.

CM Name	Size of Dataset	Leave 2 out Testing			
		R^2	MAE	MedAE	RMSE
CS1	98	0.90	0.65	0.25	1.02
CS2	89	0.90	0.71	0.59	0.86
CS3	47	0.81	2.15	2.07	2.77
CS4	90	0.83	0.72	0.88	0.90
CS5	83	0.86	0.86	0.76	1.06
CS6	46	0.72	4.06	3.15	4.99
CS7	84	0.73	1.80	0.57	2.73
CS8	50	0.86	2.52	2.13	3.24
CS9	29	0.76	2.22	1.97	2.62
CS10	57	0.87	1.04	0.86	1.26
CS11	46	0.57	0.45	0.29	0.68
CS12	43	0.81	1.12	0.82	1.34
CS13	48	0.82	5.05	4.73	5.91
CS14	103	0.76	1.62	1.36	2.16
CS15	48	0.70	1.07	0.56	1.45
CS16	35	0.11	0.61	0.45	0.84
CS17	75	0.93	2.94	2.4	3.99
CS18	79	0.53	0.91	0.44	1.21
CS19	79	0.65	1.02	0.72	1.23
CS20	68	0.82	1.83	1.65	2.32
CS21	53	0.76	10.39	8.40	12.74
CS22	97	0.81	0.32	0.18	0.47
CS23	75	0.77	0.39	0.23	0.55
CS24	54	0.71	0.54	0.43	0.66
Average	57	0.75	1.87	1.49	2.37

Abbreviations: CS – chromatographic system; MedAE - Median Absolute Error, MAE - Mean Absolute Error, and RMSE - Root Mean Squared Error

yields better predictions”.

Within the CM datasets that were collected from PredRet [7] (Table 1), both HILIC (7) and RP (13) datasets performed very well with RT-Pred. Specifically, the RP datasets had R^2 values ranging from 0.85 to 0.99 for the training sets and R^2 values ranging from 0.86 to 0.98 for the testing sets, with MAEs of 0.15 to 1.31 min and 0.18 to 1.66 min for the training and testing sets, respectively. HILIC datasets also yielded strong results with training and hold-out test sets having R^2 ranging from 0.89 to 0.98 (for training) and 0.80 to 0.97 (for testing), respectively and MAEs of 0.16 to 0.85 min (for training) and 0.17 to 1.11 min (for testing), respectively. Our results demonstrate that RT-Pred is applicable to diverse chromatographic techniques (HILIC and RP) which should make the RT-Pred server useful for a broad spectrum of users. The

unexpected success of RT-Pred handling both RP and HILIC could be attributed to the complementary nature of HILIC and RP chromatography, as highlighted in other published studies [43,58]. These studies demonstrated a similar reverse elution order for peaks identified through the two techniques. While our study primarily focused on isocratic and linear gradient RP and HILIC separation modes, the model is not inherently limited to these CMs. Given sufficient high-quality training data, we believe RT-Pred can be extended to other separation techniques such as non-linear or step gradients, strong cation exchange (SCX) or strong anion exchange (SAX). However, performance in such modes would depend on dataset availability and size.

For the smaller plant bioactive compounds datasets (24 CMs in Table 2), we incorporated L2O testing and generated 10 models per CM for evaluation. The performance of RT-Pred for the training and testing datasets was evaluated and is shown on Table 2. The average dataset size was smaller (57 RT values) yielding a lower average R^2 (0.75), a higher MAE (1.87 mins), a higher MedAE (1.49 mins), and a higher RMSE (2.37 mins). With these 24 plant bioactive datasets, it is clear that RT-Pred exhibited poorer performance compared to the performance seen with larger RP and HILIC datasets discussed above. Nevertheless, these data also show that RT-Pred is able to generate reasonably good RT prediction models (via TL) with only a few dozen RT training examples.

3.3. Performance comparison

Using five randomly selected CMs from both RP and HILIC datasets (Tables 3 and 4), we evaluated RT-Pred’s performance against other well-known ML-based RT predictors including: 1) GNN-RT [20], 2) RT-Transformer using only retained molecules and 3) RT-Transformer using all molecules [19], 4) MolCLR Fine Tuned (GIN) [21] and 5) KAN-Basic (Regression via PyKan) [22]. As these models have publicly available source code, we downloaded the programs locally, installed them and evaluated them against RT-Pred using R^2 and MAE (Table 3) as measures of performance. As seen in these tables, RT-Pred consistently outperformed all predictors across nearly all tested CMs.

We also tested RT-Pred’s performance against PredRet [7], the most widely used (non-ML) platform for RT prediction, using four randomly selected plant bioactive compounds datasets (Table 4). We scored performance using failure rates and R^2 because PredRet could not generate any RT values for a surprisingly large number of compounds. As seen in Table 4 and Fig. 4, PredRet showed a high failure rate (up to 25.3%) and poor R^2 scores (for example -0.37 for CS1 and -1.05 for CS2). In contrast, no failures were reported for RT-Pred’s predicted RTs and higher R^2 (~0.90) values were seen. These findings show that RT-Pred is both

Table 3

Comparison between different ML-based RT predicting models, including RT-Pred, on held-out test sets for a random sample of different chromatographic methods (CMs), using Mean Absolute Error (MAE) given in minutes and coefficient of determination (R^2).

RT Predictor	Chromatographic methods													
	Ajs_UoB		Fem_Long		IPB_Halle		Life_Old		Life_New		RIKEN		Eawag_XBridgeC18	
	MAE	R^2	MAE	R^2	MAE	R^2	MAE	R^2	MAE	R^2	MAE	R^2	MAE	R^2
GNN-RT	0.72	0.88	3.92	0.82	0.49	0.81	0.29	0.85	0.49	0.71	0.53	0.64	1.88	0.59
RT-Transformer (Retained molecules)	1.02	0.68	3.07	0.93	0.81	0.64	0.19	0.87	0.34	0.91	0.62	0.62	1.22	0.82
RT-Transformer (All molecules)	0.87	0.77	2.94	0.91	0.47	0.86	0.22	0.86	0.37	0.81	0.54	0.52	1.16	0.81
MolCLR Fine Tuned (GIN)	2.05	0.38	4.90	0.64	1.99	-0.33	1.67	0.24	1.05	-0.72	3.49	-2.89	4.85	-0.63
KAN-Basic	0.29	0.90	4.07	0.79	1.05	0.81	0.34	0.50	0.36	0.78	1.35	0.45	1.62	0.52
RT-Pred	0.17*	0.97	1.66	0.98*	0.48	0.93	0.18	0.92	0.34	0.88	0.40	0.91	0.90	0.90

* The lowest MAE and the highest R^2 are bolded.

Table 4

Comparison of PredRet's performance to RT-Pred's performance across four randomly chosen chromatographic methods (CMs) related to plant bioactive compounds by failure rate (scored as percentages) and coefficient of determination (R^2).

CM Name	Failure Rate (%)		R^2 score	
	PredRet	RT-Pred	PredRet	RT-Pred
CS1	25.3%	0%	-0.37	0.90
CS2	13.1%	0%	-1.05	0.90
CS4	18.4%	0%	0.19	0.83
CS5	19.3%	0%	0.56	0.86

Abbreviation: CS – chromatographic system.

accurate and robust in its ability to predict RTs over a wide range of CMs and even in situations where the CM training set is unusually small.

Furthermore, to ensure a rigorous comparison, we also directly evaluated RT-Pred on the exact HILIC and RPLC datasets used in Retip's publication, which were downloaded from their Supplementary Materials. For the HILIC dataset (970 compounds), RT-Pred achieved MAEs of 0.50 min (training) and 0.65 min (testing), surpassing Retip's reported MAEs of 0.70 min (training) and 0.78 min (testing). Similarly, for the RPLC dataset (RIKEN PlaSMA – 494 compounds), RT-Pred achieved MAEs of 0.35 min (training) and 0.36 min (testing), outperforming Retip's MAEs of 0.50 min (training) and 0.57 min (testing).

3.4. Prediction of RTs with RT-Pred for multiple compound databases

To assess the ability of RT-Pred to handle a wide range of chemical types and to robustly predict RTs over many different CMs, we ran RT-Pred (using 20 different CMs) on three large and diverse chemical datasets maintained in our laboratory. These include the Human Metabolome Database (HMDB), with over 220,000 human metabolite entries [24], FooDB, a food composition database with more than 80,000 food compound entries [25] and NP-MRD, a database containing >280,000 natural products from plants, microbes and animals [26]. Each of these databases encompasses a wide range of chemical structures, including both water-soluble and lipid-soluble compounds. The rate at which RT-Pred could predict RTs for the compounds in these databases (for any given CM) was approximately ~8000 compounds/minute on a Linux Server with an Intel Xeon CPU rated at 2.20 GHz and 4 cores. In total, 13 million RTs were predicted covering 20 CMs and 535,848 non-redundant molecules. No failures were reported in any of the RT predictions, which highlights both the robustness of RT-Pred and the high quality of the structures in HMDB, FooDB and NP-MRD. These predicted RTs for each of the different databases and for each of the CMs are available for download on the RT-Pred website under the "Download" tab. Users can also search through the RT-Pred's collection of molecule-specific and CM-specific RTs through the various search functions offered by the RT-Pred webserver (see below). Overall, the application of RT-Pred to the thousands of compounds found in

HMDB, FooDB and NP-MRD should prove to be quite helpful for many researchers in metabolomics, food/nutrition science and natural product chemistry.

3.5. The RT-Pred webserver

The RT-Pred webserver is available at <https://rtpred.ca>. It supports five different functions: 1) predicting RTs for multiple query molecules; 2) creating custom RT predictors; 3) predicting RTs for multiple query molecules using a custom RT predictor; 4) searching for RTs of known molecules; 5) searching for molecules with specific RTs. The latter two functions exploit RT-Pred's large database of pre-calculated RTs generated over multiple pretrained CMs for molecules in the HMDB, FooDB and NP-MRD [24–26]. The workflow of the first two functions for predicting RTs and creating predictions are illustrated as flowcharts in Fig. 5. RT-Pred also has a CM database containing comprehensive information about each of the CMs used to create and test RT-Pred. Furthermore, users can easily deposit their own custom CMs, or their own sets of RT data generated via their custom CMs, to share with other RT-Pred users for future research.

A screenshot of the RT-Pred server homepage is shown Fig. 6A. The most commonly used functions "Predict retention time", "Build custom retention time predictor" and "Search retention time for compounds" can be accessed by clicking on any one of the three sliding bars found in the center of the home page. These functions can also be accessed via tabs on RT-Pred's menu bar (located at the top of the homepage). The menu bar displays nine different tabs for nine different RT-Pred modules: 'Instructions', 'Predict RT', 'Build Custom RT Predictor', 'Search RT Data', 'Deposit RT Data', 'Download RT Data', 'CM Database', 'About', and 'Contact Us' (Fig. 6A). Clicking on the 'Instructions' tab allows users to view three different, richly annotated and fully illustrated tutorials on how to use RT-Pred and how to create a custom-built RT predictor. Clicking on the 'Predict RT' displays a flowchart to guide users through the prediction process. A link to an InChI to SMILES converter can be found here (which is also available under the 'About' menu). Within the 'Predict RT' module, users have the option to predict RTs from a list of different CMs including multiple RP or HILIC systems. After selecting the specific CM, users can then upload a *.csv file with a list of molecules in SMILES format (up to ~7000). First-time users can also pre-load an example by clicking on 'Load Example' button, and then press 'Submit' to initiate the RT predictions for the chosen CM (Fig. 6B). After the predictions are completed, users can review the predicted RTs on a newly generated webpage and download the RT/structure file as a *.csv file by clicking on the 'Download .csv' button located at the top of that page (Fig. 6C).

Clicking on the 'Build Custom RT Predictor' from the menu bar also displays a step-by-step flowchart. To operate this function, users must first provide information about the dataset by choosing from the drop-down options for the 'Molecular Structure Notation', 'Retention Time Unit' and 'Column Type' used during the collection of the dataset. User must also provide detailed CM information, which includes:

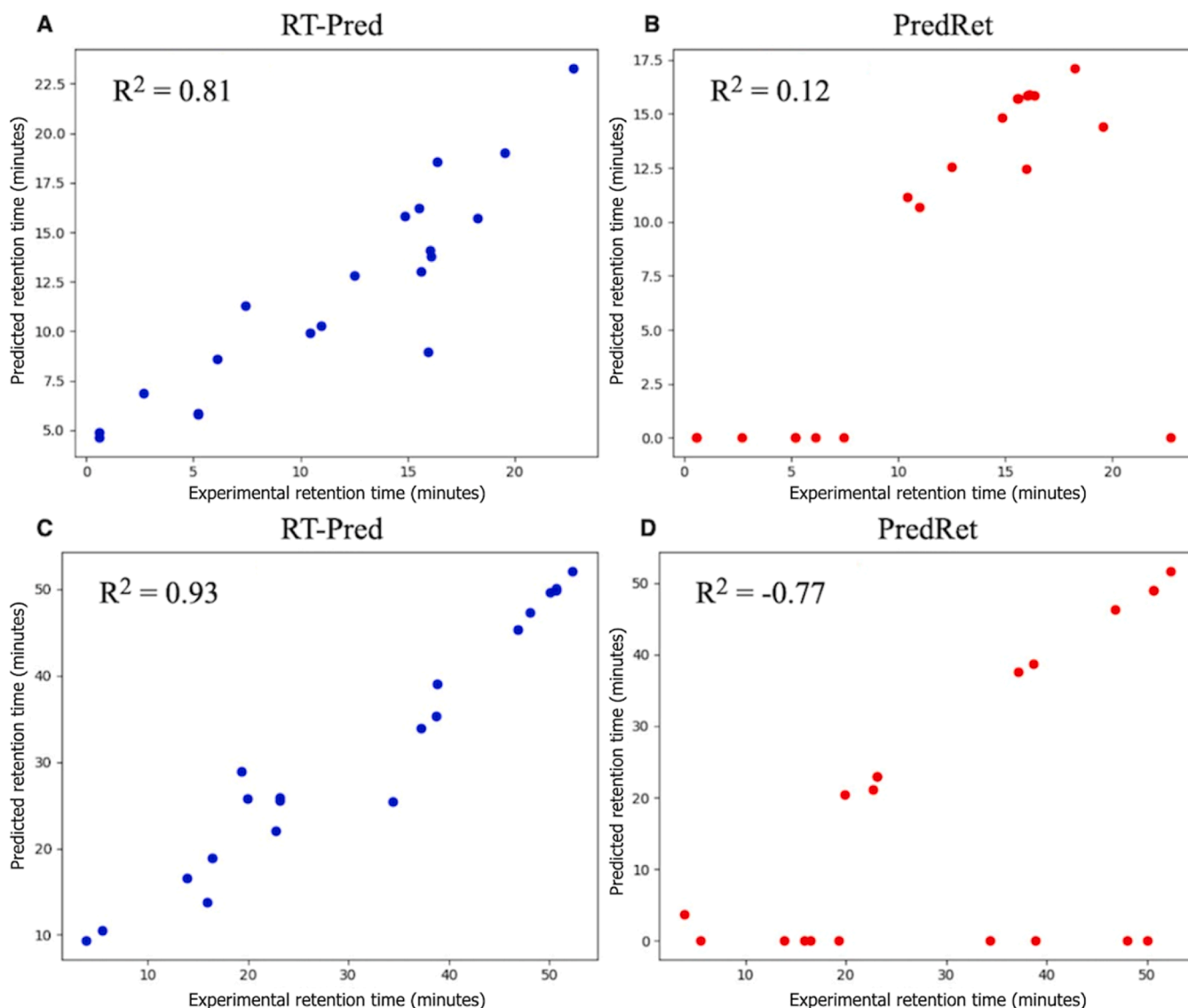


Fig. 4. Correlation plots showing the predicted versus experimental retention times (RTs) predicted by RT-Pred (A, C) compared to PredRet (B, D) using plant bioactive compounds in two different CMs, namely chromatographic system 3 or CS3 (A, B) and CS17 (C, D). Due to limited data size, the same 20 test molecules withheld during RT-Pred's L2O test modeling were included in the evaluation for validity. Notably, RT-Pred (blue, A, C) consistently outperformed PredRet (red, B, D), demonstrating superior accuracy, efficiency, and generalizability (higher R^2 values).

'Chromatographic Method Name', 'Column', 'Eluent', 'Eluent Additives', 'Eluent pH', and optional details (such as the depositor's name, affiliation and email) in the 'Reference' and 'Meta Data' sections. After filling in the CM and meta-data information, users must upload a *.csv file with their list of molecular structures (in SMILES format) and measured RTs for their chosen CM (Fig. 6D). First-time users can also pre-load an example by clicking on 'Load Example' to go through the process of building a custom RT model. After clicking 'Submit', the dataset is submitted, and the users are notified about their CM number and prompted to click on 'Submit' to begin the custom-RT predictor model training. In the next page, the users are informed of the approximate training time (usually less than 15 mins), and the page automatically reloads after model training is finished. Performance statistics (R^2 , MAE, etc.) are also provided here so that users can assess the quality and utility of their custom RT predictor. If the users are not satisfied with the performance, they can click on 'No' to be redirected back to the 'Build Custom RT Predictor' page. If satisfied with the performance, the users can click on 'Yes' after which their intent to optionally donate their custom-RT predictor model is asked. To answer, the users can either

select 'Yes' or 'No' and both options will lead them to the prediction page for their custom-RT predictor where they may then upload a set of up to 7000 SMILES strings to have their RTs calculated for the uploaded structures. The RT prediction with the custom predictor can be done in a manner identical to the 'Predict RT' function described earlier. Fig. 6C shows a screenshot of the predicted RTs after an input file containing SMILES was uploaded for a custom CM and the 'Submit' button was clicked. Before making any RT predictions, users can optionally choose to save their model in the RT-Pred server by clicking on 'Save model'. Alternatively, users can click 'Delete model' if they do not wish to have others use their custom RT-predictor. This will permanently remove all model data, and all information associated with the CM and its CM-specific RT predictor from the server. However, for users who wish to share their models publicly, RT-Pred allows submission for inclusion in a new CM category. Before making these models publicly and permanently available, the RT-Pred team tests them for accuracy, relevance, and completeness to ensure quality and prevent misuse or abuse.

Another feature of the RT-Pred server (via the 'Deposit RT Data' module) is its ability to collect experimental RT data from users (who do

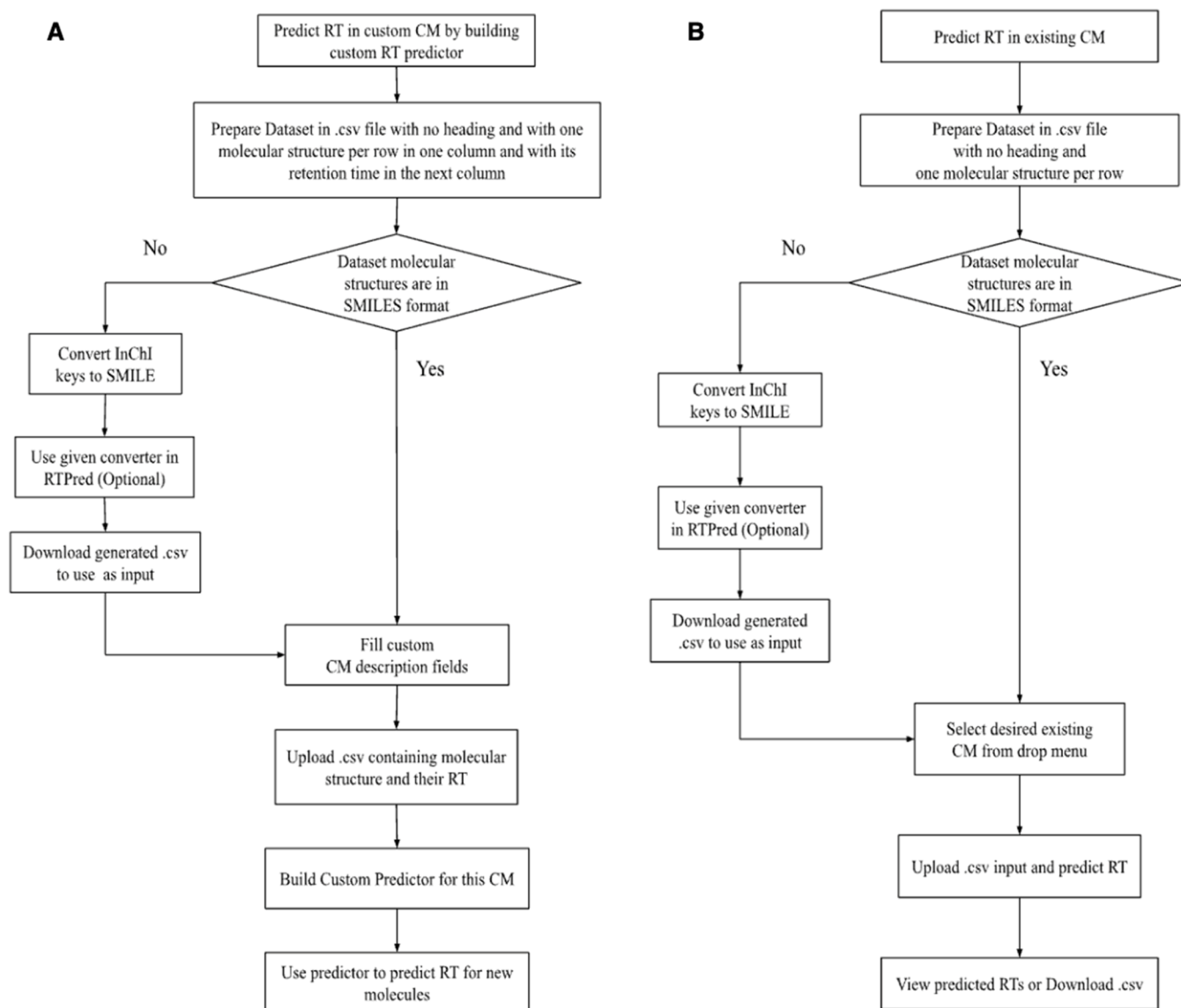


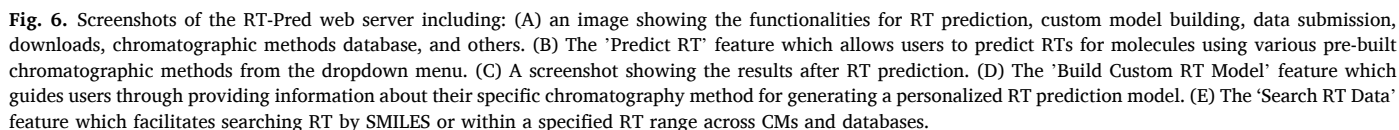
Fig. 5. Flow charts showing (A) the steps RT-Pred uses for predicting RTs of molecules in existing chromatographic methods (CMs) in the server and (B) the process RT-Pred employs to construct a customized RT prediction model for molecules within a user-defined CM which can be later used for further predictions.

not need to make a custom predictor but wish to make their data available to others). This option is available so that other researchers can use that deposited RT data to build custom RT predictor models or for other purposes. For this, clicking on the ‘Deposit RT data’ displays a form (similar to the ‘Build Custom RT Predictor’ form) where a new RT dataset and new CM descriptions can be added and deposited to the RT-Pred server for use by other users at a later time. The deposited data may also be used by the RT-Pred team which (for higher quality CM models) will be used to calculate RTs for all compounds in the NP-MRD, HMDB and FooDB and added to the RT-Pred database [24–26].

RT-Pred also supports several database search functions. Clicking on the ‘Search RT Data’ tab from the webserver’s menu bar allows users to search RT-Pred’s databases via two approaches. The first (Compound-to-RT) allows users to search RT-Pred’s databases by compound (via an entered SMILES string) to find an RT for that compound measured/predicted using a specified CM. The second (RT-to-Compound) allows users to search for compounds via an entered RT range for a specific CM and to see what compounds in RT-Pred’s databases are identified (Fig. 6E). For the Compound-to-RT search, users must first select a specific CM from a pull-down menu and enter a SMILES string (for a specific molecule). Then after clicking the ‘Submit’ button, the RT of the

molecule for the specified CM is displayed. For the RT-to-Compound search, users must enter an RT range, select a CM and after clicking the ‘Submit’ button, a list of matching compounds and their predicted RTs is displayed on a new webpage. This set of matching compounds and RTs can then be downloaded as a *.csv file by clicking on the ‘Download’ button at the bottom of the page. As always, first time users can also click on the ‘Load Example’ button to become more familiar with how the ‘Search RT Data’ function works.

A number of other useful functions in RT-Pred are also available on the menu bar. For instance, clicking on the ‘Download RT’ tab from the menu bar displays files with hyperlinks to the predicted RTs generated for the HMDB, FooDB and NP-MRD databases for all high performing CMs. Clicking on the ‘CM Database’ displays detailed descriptions of all supported CMs in the RT-Pred webserver. Clicking on the ‘About’ tab generates a brief overview of the RT-Pred server, along with information about the webserver’s operating system and web browser compatibility, more detailed explanations of RT-Pred algorithms, how to cite the RT-Pred webserver, summary statistics of the performance of RT-Pred for all uploaded CMs and a link to the InChI to SMILES converter tool. Clicking on “Contact us” displays an information page detailing how users can contact the RT-Pred team. Clicking on the “Question/Feedback



4. Conclusion

We have described an easy-to-use RT prediction webserver (called RT-Pred) that predicts RTs for small molecules for almost any user-specified CM. This webserver was developed to address a number of

gaps in RT prediction and RT prediction support. These gaps include a preponderance of difficult-to-use software, poor support for custom RT predictions, limited RT database accessibility, low quality RT predictions, and an inability to distinguish normally eluting molecules from void-volume and retained molecules. Consequently, we designed the RT-Pred program and the RT-Pred webserver to address these limitations.

Among the key innovations and functional offerings associated with RT-Pred (and its webserver) are: 1) the development of the most accurate RT predictor described to date; 2) the implementation of a void, retained and eluted classifier (to improve the accuracy of RT predictions); 3) the creation of an open access webserver for predicting RTs over many different CMs; 4) the development of an open access webserver for training and creating custom RT predictors for specific CMs; 5) the implementation of an open access, custom RT prediction/model deposition system; 6) the generation of a large, searchable and downloadable database of millions of predicted RTs for dozens of different CMs and 7) the assembly of a downloadable database of thousands of experimentally measured RTs for >40 different CMs.

To provide as much transparency to the process and methods associated with RT-Pred, we have described in detail how RT-Pred was trained and validated. We have also provided extensive information about how RT-Pred's algorithms were developed and tested as well as how the predicted set of RTs (and their corresponding CMs) for >500,000 known metabolites and natural products were generated. Similarly, the processes by which the webserver functions have all been explained with tutorials and illustrated workflows (available on the RT-Pred website). Likewise, all of the datasets used in training and validating the RT-Pred software as well as all the source code for running the programs are available on request.

RT-Pred offers many unique features and exhibits exceptional accuracy and performance. However, it is important to be aware that it does have some limitations. Specifically, those RTs predicted for smaller CM training datasets (<200 training examples), such as the plant bioactive compounds, will exhibit relatively poorer performance than those trained on larger training datasets (>200 training examples). Therefore, it is important to be aware of these training size limitations, both with the RT database in RT-Pred and in developing custom RT predictors. Likewise, the RT-group classifier (RTGC), while effective in identifying void and retained molecules in large datasets, was unable to classify compounds from smaller (<200 examples) CM datasets due to their very limited data size. Expanding the RTGC classification tool to handle all datasets, no matter the size, could further enhance RT-Pred's reliability. However, we would strongly encourage anyone trying to create CM-specific RT dataset for creating a custom RT predictor to use >200 molecules.

Despite being initially trained on an RP training set, the ability of RT-Pred to handle HILIC data via transfer learning was a surprise. However, upon further reflection, the fact that HILIC elution patterns are often the reverse of RP elution patterns [43,58] and the fact that retention times for both CMs depend on the same physical properties of the eluents (but with different coefficients), it is not unreasonable to expect the transfer learning process to be able to adjust or invert the model weights and produce a model yielding high quality retention times for HILIC data. This is precisely what happened here, and it aligns with similar findings for other transfer learning models aimed at, for instance, upward/downward stock market trend prediction. It is also important to note that RT-Pred's performance may vary when applied to non-linear or step gradients, isocratic methods or separation techniques like SAX and SCX, as their retention mechanisms differ from those of linear gradient RP and HILIC, which were the primary focus of RT-Pred's training.

Looking forward, we expect to continue to add more features to RT-Pred over the coming years. These include expanding RT-Pred's CM coverage to incorporate a broader range of CMs, re-training some of the smaller CM models with more RT training examples, speeding up the training process using GPU accelerators, expanding RT-Pred's database

of predicted RTs to include more chemical exposure and drug compounds (i.e., the exposome) and, finally, improving RT-Pred's data conversion protocols to handle molecular structures more consistently and with fewer errors. These improvements will allow RT-Pred to support a wider variety of applications in metabolomics, foodomics, exposomics, natural product chemistry and analytical chemistry.

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CRediT authorship contribution statement

Mahi Zakir: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis, Data curation. **Marcia A. LeVatte:** Writing – review & editing, Writing – original draft, Investigation. **David S. Wishart:** Conceptualization, Funding acquisition, Methodology, Supervision, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.chroma.2025.465816](https://doi.org/10.1016/j.chroma.2025.465816).

Data availability

Most data is available for download from the webserver. Additional data is available upon request.

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